IMU-Based Step Counting

Causal TCN Event Detection and Real-Time Deployment

Mio Zhu

Department of Electrical and Electronic Engineering
The University of Hong Kong

Roadmap

1. Problem
2. Data
3. Pipeline
4. Model
5. Training
6. Inference
7. Evaluation
8. Deployment

Goal and Constraints

Task

Count walking steps from IMU recordings with one interface for offline evaluation and real-time updates.

Inputs

- Required: time and 3-axis acceleration
- Optional: gyro and magnetometer
- Irregular raw sample rates handled by resampling

Outputs

- Offline: step count and timestamps
- Streaming: new steps and cumulative total
- Diagnostics for threshold and model state

Design challenge

Detect sparse step events while staying robust to phone orientation, gait speed, sensor placement, and chunk boundaries.

Dataset and Ground Truth

- Training data: OxWalk Dec 2022 IMU recordings
- 39 participants used in the final retrain
- Four modalities: hip/wrist at 100 Hz and 25 Hz
- Participant-level split prevents the same person appearing in train and test

Split	Participants	Windows
Train	23	649,794
Val	8	210,000
Test	8	230,164

Label construction

Binary step annotations are resampled to 50 Hz, then widened with a Gaussian label of $\sigma = 0.06$ s for event learning.

Preprocessing and Windowing

1. Parse timestamped accelerometer CSV files and convert timestamps to seconds.
2. Resample every recording to a common 50 Hz timeline with linear interpolation.
3. Build an orientation-aware feature vector:

$$x_t = [a_x(t), a_y(t), a_z(t), \|a(t)\|_2]$$

4. Slice into 4 s windows with 0.5 s stride, giving 200 samples per window.
5. Standardize features using train-set mean and standard deviation saved in the checkpoint.

Why include acceleration magnitude?

The raw axes preserve motion direction, while $\|a\|_2$ gives a stable cue when phone orientation changes.

Causal TCN StepNet Architecture

Backbone

- Input projection: 4 channels $\rightarrow$ 64 channels
- 4 causal residual convolution blocks
- Kernel size 5, dilations $1, 2, 4, 8$
- BatchNorm, GELU, dropout 0.15

Two heads

- Event head: per-sample step probability
- Count head: expected number of steps in the window
- Causal convolutions support streaming inference

Acc window $\rightarrow$ feature normalization $\rightarrow$ causal TCN $\rightarrow$ event logits + count estimate

Model Comparison

Method	Strengths	Observed limitations
Peak heuristic	Simple, fast, interpretable; no training required	Sensitive to thresholds and sensor placement; subset MAE 15.6 steps
LSTM StepNet	Learns temporal patterns and gait rhythm	Sequential recurrence is slower to train/infer; no final checkpoint retained
Causal TCN StepNet	Parallel training, fixed streaming context, event + count heads	Needs labeled data and probability threshold calibration

Final choice

TCN was selected because causal convolutions preserve live behavior while training efficiently on 649,794 windows. Final test count MAE: 0.484 steps/window.

Training Objective and Hyperparameters

Loss

$$L = L_{BCE}(y_{event}, \hat{y}_{event}) + 0.15 L_{MSE}(y_{count}, \hat{y}_{count})$$

Positive event samples use weight 3.0 to counter sparse labels.

- Optimizer: AdamW
- Learning rate: 10^{-3}
- Weight decay: 10^{-4}
- Batch size: 256
- Early stopping patience: 6 epochs
- Best epoch: 4

Checkpoint contents

The saved model stores network weights, feature normalization statistics, sample rate, window size, split metadata, and seed.

Offline and Real-Time Inference

Offline path

- Validate time and acceleration shape
- Sort timestamps and remove duplicates
- Normalize start time to zero
- Resample to model rate and call the same update logic

Streaming path

- Keep a 4 s context buffer
- Run causal TCN on context + new chunk
- Detect local maxima above probability threshold 0.68
- Enforce minimum step interval of 0.33 s

Boundary handling

Previous probabilities and timestamps are retained, so peaks crossing chunk boundaries are counted exactly once.

Experimental Results

Metric	Value
Best validation loss	0.3732
Test loss	0.4562
Test event loss	0.3364
Test count loss	0.7992
Test count MAE	0.4840

Own recordings

Trial	GT	Pred.	Error
test1	84	82	-2
test2	100	101	+1

The dedicated model-comparison slide explains why the causal TCN was selected over the heuristic and LSTM alternatives.

Real-Time Demo and Takeaways

phyphox demo

- Polls Remote Access API every 0.05 s
- Auto-detects time and acceleration buffers
- Displays acceleration curves, detected steps, and cumulative count
- Web dashboard available through Flask and Chart.js

Key lessons

- Causality matters for live feedback
- Magnitude feature improves orientation robustness
- Participant split is critical for realistic evaluation
- Threshold and minimum interval convert probabilities into stable events

Limitations and next steps

More validation is needed for non-walking motions, phone transitions, and device-specific acceleration scaling.